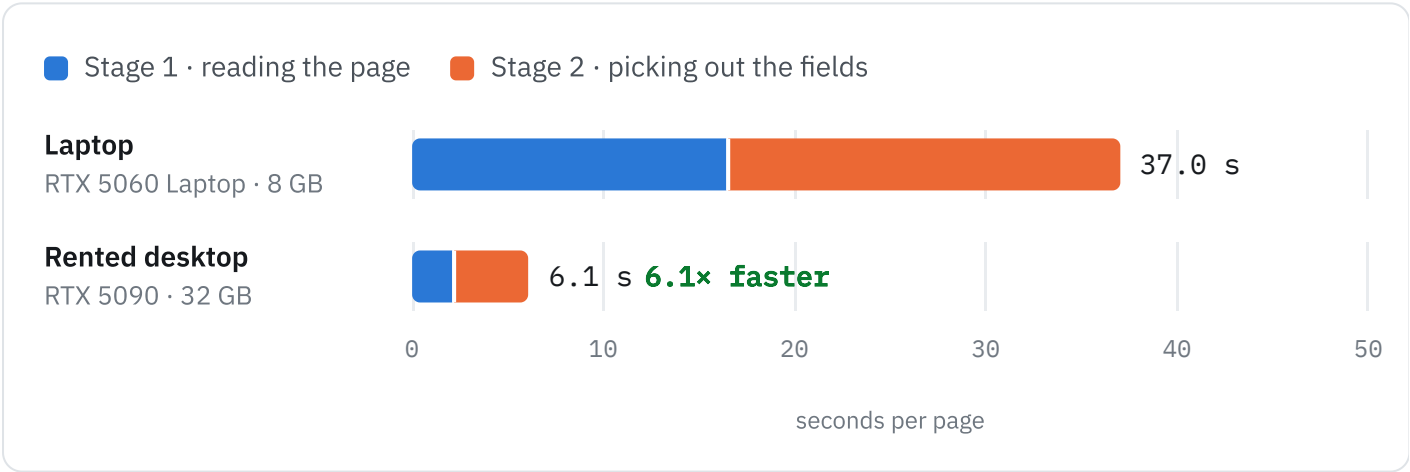


A rented RTX 5090 reads a receipt page in 6.1 s, about 6× faster than the laptop

Same 15 scanned pages, same two AI models, same settings, run one page at a time on each machine. The laptop takes 37.0 seconds per page; the rented desktop takes 6.1.

6.1s per page on the RTX 5090 laptop 37.0 s	6.1× faster end to end reading 7.8× · fields 5.3×	~590 pages per hour laptop ~97	61s for a 10-page batch laptop 370 s · limit 600 s
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Where the time goes



Side by side

	Laptop	Rented desktop	Change
Graphics card	RTX 5060 Laptop	RTX 5090	
Graphics memory	8 GB	32 GB	4×
Processor	Intel laptop CPU	Threadripper PRO 7975WX	
Stage 1 · reading	16.4 s	2.1 s	7.8× faster
Stage 2 · fields	20.5 s	3.9 s	5.3× faster
Total per page	37.0 s	6.1 s	6.1× faster
Graphics memory used	5.4 of 8 GB	10.3 of 32 GB	
Start-up (models loading)	46 s	6 s	
Pages that failed to read	0 of 15	0 of 15	

What it means for the webapp

40-page batch	The webapp's stated maximum batch. About 4 minutes on the RTX 5090 , inside the 10-minute limit for one batch, one page at a time. The laptop would need about 25 minutes, so it cannot handle a batch that size.
300-page job	About 30 minutes on the RTX 5090, against roughly 3 hours on the laptop.
Headroom	The RTX 5090 used 10 of its 32 GB. That spare memory is what would let it work on several batches at once, but that was not tested (see below).

What this test does not show

- **Accuracy.** This test only timed the machines. It did not check the answers against the answer key. The same models and settings were used, so the answers should be very close to the laptop's, but different graphics cards can round numbers slightly differently. That has not been checked on the RTX 5090.
- **Several users at once.** Pages went through one at a time. How the RTX 5090 performs with several batches arriving together is still unmeasured, and it is what sizing a shared server depends on.
- **Other document types.** All 15 pages were scanned cash bills. Toll tickets and long multi-page receipts were not in this set, and no page needed the slower second read that the pipeline does on hard pages.
- **Where the documents would live.** The test ran on a rented cloud machine using test documents only. The company prefers to keep real documents on its own hardware, so these numbers describe what an RTX 5090-class card delivers, not a proposal to process real documents in the cloud.

How it was measured. Both machines ran the same benchmark in the project's pipeline notebook: two scanned cash-bill documents (5 and 10 pages), stage 1 `typhoon-ocr1.5-3b` and stage 2 `qwen3:4b`, identical settings and pipeline code. Both models were loaded and run once before timing started, and that warm-up is not counted. Times are the average over all 15 pages. The server adds a few checks on top of these two stages; they were measured earlier at about 0.1% of the time and are left out. Both runs were on 18 September 2026; the rental was a Vast.ai instance.